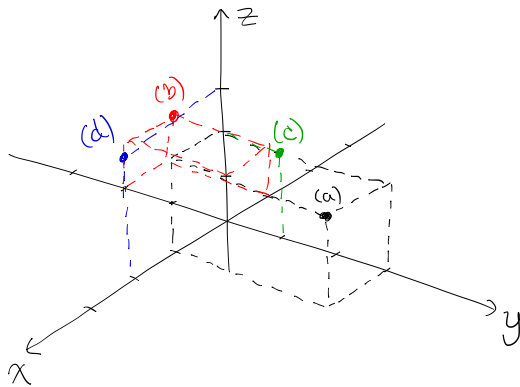


Calculus for Biology and Medicine, 3rd Edition, Claudia Neuhauser

10.1 # 3, 5, 13, 14, 19

3)



$$\begin{aligned} 5) f(2,3) &= \frac{2(2)}{(2)^2 + (3)^2} \\ &= \frac{4}{4+9} = \boxed{\frac{4}{13}} \end{aligned}$$

$$19) f(x,y) = 1+x^2+y^2 \quad \boxed{\text{Figure 10.23}}$$

$$13) f(x,y) = x^2 + y^2$$

$$\text{Domain: } \mathbb{R}^2$$

$$\text{Range: } \mathbb{R}$$

$$x^2 + y^2 = c$$

$$c=1: x^2 + y^2 = 1$$

$$c=4: x^2 + y^2 = 4$$

$$14) f(x,y) = \sqrt{9-x^2-y^2}$$

$$\text{Domain: } \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 9\}$$

$$\text{Range: } \{z \in \mathbb{R} \mid 0 \leq z \leq 3\}$$

$$\sqrt{9-x^2-y^2} = c \Rightarrow x^2 + y^2 = 9 - c^2$$

$$c=0: x^2 + y^2 = 9$$

$$c=\sqrt{5}: x^2 + y^2 = 4$$